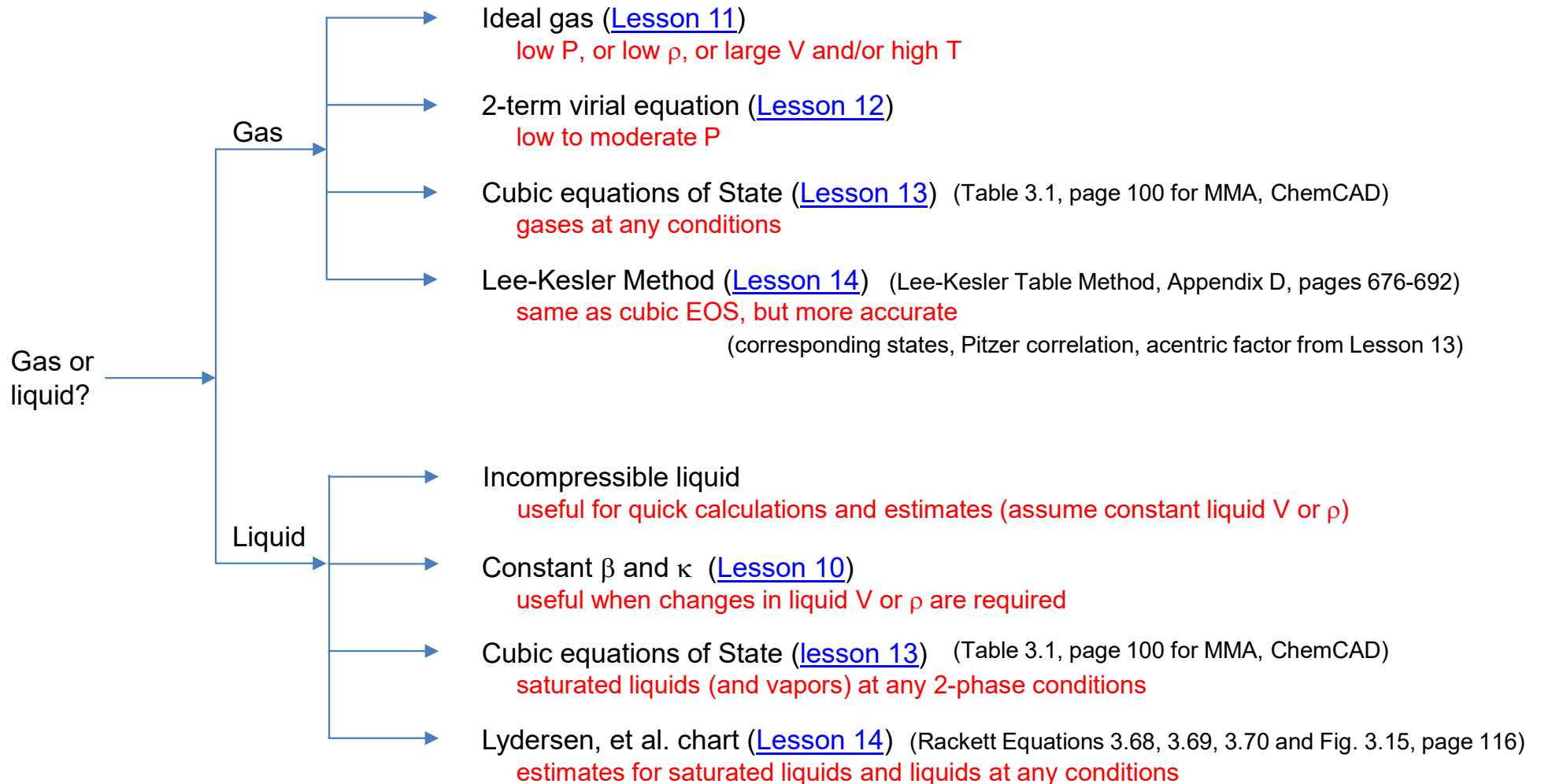


CH365 Chemical Engineering Thermodynamics

Lesson 15 Review

Equation of State Decision Tree

Be able to intelligently select an appropriate equation of state or generalized correlation for a given application.



IMPORTANT: Be able to apply cubic equations of state to calculation of work using $\int P dV$ or $\Delta(PV)$ given $P(V)$ from equation 3.41

Homework

2nd Virial Coefficients

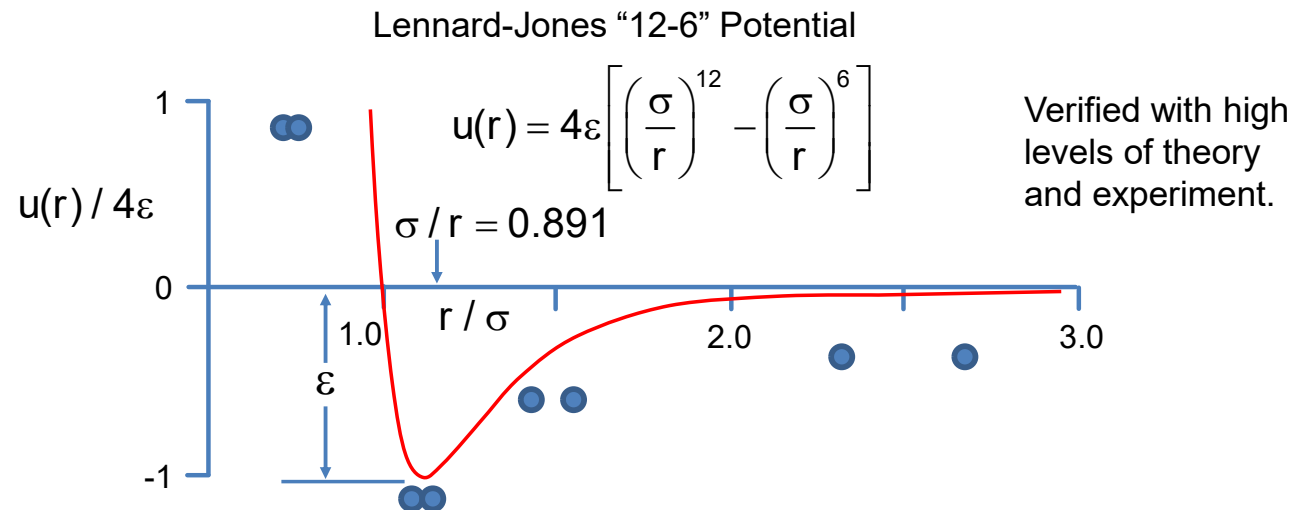
Slide 4

Can be Derived from Theory

2-molecule pair-wise IMFs



Electrostatic forces
(H atom + H atom → H₂)
London dispersion forces
(He, Ne, Ar, Kr, Xe)



B_2 is found by
incorporating $u(r)$ into a
Boltzmann distribution
and integrating the
resulting distribution
over all r .

Second Virial Coefficient:

$$B_2 = B_2(T) = 2\pi \int_0^{\infty} \left(1 - \exp \left\{ -\frac{4\epsilon}{kT} \cdot \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] \right\} \right) r^2 dr$$

“An Introduction to Applied Statistical Thermodynamics,” Stanley I. Sandler,
Wiley: 2010, pp. 130-131.

Similar theory exists for each of the other virial coefficients.

2nd Virial Coefficients

Slide 5

Can be Derived from Theory

2-molecule pair-wise IMFs



Electrostatic forces
(H atom + H atom $\rightarrow$ H₂)

London dispersion forces
(He, Ne, Ar, Kr, Xe)

Problem 3.78

The *Boyle temperature* is the temperature for which:

$$\lim_{P \rightarrow 0} \left(\frac{\partial Z}{\partial P} \right)_T \equiv 0$$

- (a) Show that the second virial coefficient B is zero at the Boyle temperature.
- (b) Use the generalized correlation for B , Eqs. 3.58 to 3.62, to estimate the reduced Boyle temperature for simple fluids.